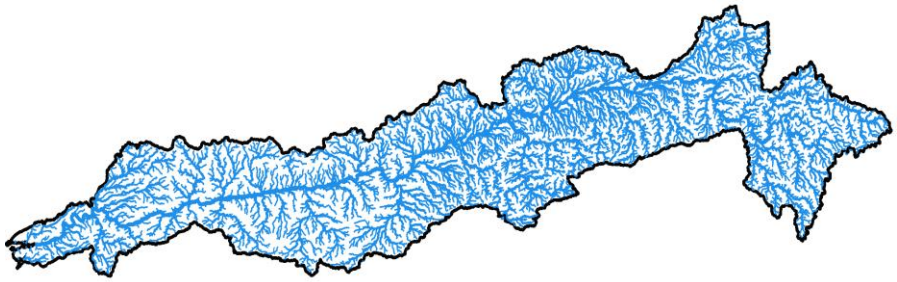
National River Conservation Directorate
Ministry of Jal Shakti, Department of Water Resources,
River Development & Ganga Rejuvenation
Government of India

Identification of the Major Threats and Challenges for the Narmada River



March 2026

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National River Conservation Directorate (NRCD)

The National River Conservation Directorate, functioning under the Department of Water Resources, River Development & Ganga Rejuvenation, and the Ministry of Jal Shakti, provides financial assistance to the State Government for the conservation of rivers under the Centrally Sponsored Schemes of 'National River Conservation Plan (NRCP)'. National River Conservation Plan to the State Governments/ local bodies to set up infrastructure for pollution abatement of rivers in identified polluted river stretches based on proposals received from the State Governments/ local bodies. (www.nrcd.nic.in)

Centers for Narmada River Basin Management and Studies (cNarmada)

The Center for Narmada River Basin Management and Studies (cNarmada) is a Brain Trust dedicated to River Science and River Basin Management. Established in 2024 by IIT Gandhinagar and IIT Indore, under the supervision of cGanga at IIT Kanpur, the center serves as a knowledge wing of the National River Conservation Directorate (NRCD). cNarmada is committed to restoring and conserving the Narmada River and its resources through the collation of information and knowledge, research and development, planning, monitoring, education, advocacy, and stakeholder engagement. (www.cnarmada.org)

Centers for Ganga River Basin Management and Studies (cGanga)

cGanga is a think tank formed under the aegis of NMCG, and one of its stated objectives is to make India a world leader in river and water science. The Centre is headquartered at IIT Kanpur and has representation from most leading science and technological institutes of the country. cGanga's mandate is to serve as a think-tank in the implementation and dynamic evolution of the Ganga River Basin Management Plan (GRBMP) prepared by the Consortium of 7 IITs. In addition, it is responsible for introducing new technologies, innovations, and solutions into India. (www.cganga.org)

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This report is a comprehensive outcome of the project jointly executed by IIT Gandhinagar (Lead Institute) and IIT Indore (Fellow Institute) under the supervision of cGanga at IIT Kanpur. It was submitted to the National River Conservation Directorate (NRCD) in 2026. We gratefully acknowledge the individuals who provided information for this report.

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Preface

The Narmada River, often referred to as the lifeline of central and western India, holds unparalleled significance for the region's water resources, ecosystems, and communities. Its hydrological complexity, combined with its ecological and cultural heritage, underscores the critical need for its preservation and sustainable management.

This report focuses on identifying the major threats and challenges in the Narmada River, which directly and indirectly affect its health, including water quality and ecological health. The collection and compilation of data of the Narmada Basin will provide valuable insights into the Narmada Basin.

The findings and recommendations presented in this report are intended to serve as a guiding framework for policymakers, researchers, and practitioners committed to sustainable management of these nutrients and sediments in the Narmada Basin. By identifying trends and challenges, it aims to support policy development, thereby facilitating ecological preservation alongside socioeconomic progress in the Narmada River Basin.

We are deeply grateful to all individuals, organizations, and institutions that contributed to the preparation of this report. Their dedication, expertise, and support have been instrumental in shaping this study. This report will inspire collaborative efforts and meaningful action toward the long-term health and sustainability of the Narmada River and its Basin.

Centre for Narmada River Basin Management and Studies (cNarmada)
IIT Gandhinagar, IIT Indore

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Abbreviations & Accronyms

BCM	Billion Cubic Meters
BOD	Biochemical Oxygen Demand
CETP	Common Effluent Treatment Plants
CGWB	Central Ground Water Board
CIFRI	Central Inland Fisheries Research Institute
CPCB	Central Pollution Control Board
CWC	Central Water Commission
DO	Dissolved Oxygen
ETP	Effluent Treatment Plants
F. Coliform	Faecal Coliform
GPCB	Gujarat Pollution Control Board
GRBMP	Ganga River Basin Management Plan
MLD	Million Liters per Day
MoDWS	Ministry of Drinking Water and Sanitation
MPN	Most Probable Number
MPPCB	Madhya Pradesh Pollution Control Board
NRCD	National River Conservation Directorate
NRCP	National River Conservation Plan
NWDA	National Water Development Agency
ODF	Open Defecation Free
SBM	Swachh Bharat Mission
STP	Sewage Treatment Plant
T. Coliform	Total Coliform
TDS	Total Dissolved Solids
UNICEF	United Nations International Children's Emergency Fund
WHO	World Health Organization

1. Overview of the Narmada Basin

India's diverse landscapes are defined by its magnificent river systems, which originate in the Himalayas and flow into the Western and Eastern Ghats. The Narmada River ranks as the fifth-longest river in India and is the longest river flowing westward. The Narmada traverses the states of Madhya Pradesh and Gujarat and is often referred to as the "Lifeline of Madhya Pradesh and Gujarat" for its substantial contributions to both states. Originating from the Amarkantak Plateau in the Anuppur district of Madhya Pradesh, the Narmada forms the traditional boundary between North and South India. It flows westward for 1,312 km before emptying into the Arabian Sea through the Gulf of Khambhat, situated 30 km west of Bharuch city in Gujarat. The Narmada basin extends from 21° 40' 12'' to 23° 41' 24'' N latitudes and 72° 48' 36'' to 81° 45' 36'' E longitudes, covering a total area of 98796 sq. km, which is nearly 3% of the total geographical area of India (3,297,427.32 sq. km). The upper part of the river basin primarily comprises hilly regions, whereas the middle and lower reaches are fertile and broad, making them well-suited for cultivation. The basin's annual water potential is 45.65 billion cubic meters (BCM), with a utilizable water potential of 34.50 BCM (75.57%). The entire stretch of the River Narmada can be primarily divided into three sub-basins/segments based on their geomorphology, ecology, and rheology

Upper Narmada sub-basin: ≈ 720 km Amarkantak to Hoshangabad

Middle Narmada sub-basin: ≈ 485 km Hoshangabad to Navagam

Lower Narmada sub-basin: ≈ 145 km Navagam to Gulf of Khambhat

The Narmada River drainage network comprises 19 major tributaries (a total of 41). According to the 2011 Census, the basin's total population is 61,243,103, with 32,396,859 males (52.89%) and 30,264,244 females (47.11%). The Narmada flows through a rift valley, surrounded by the Vindhyas to the north, the Maikala range to the east, the Satpuras to the south, and the Arabian Sea to the west. It lies at the northern extremity of the Deccan plateau and covers major portions of the states of Madhya Pradesh and Gujarat, along with smaller parts of Chhattisgarh and Maharashtra. The Narmada River travels through Madhya Pradesh (1,077 km), Maharashtra (74 km), including along the border between Madhya Pradesh and Maharashtra (39 km), then the border between Maharashtra and Gujarat (74 km), and finally Gujarat (161 km).4. The basin spreads broadly over 40 districts, comprising 27 districts of Madhya Pradesh, seven districts of Gujarat, four districts of Chhattisgarh, and two districts of Maharashtra (Narmada Basin, 2014; River Atlas, 2025)

2. Introduction

Rivers play a vital role in human civilization, serving as essential lifelines for communities around the globe. They are rich in biodiversity, and their clean water is crucial for the survival and well-being of countless species, including humans. In addition, rivers are key contributors to urbanization and industrialization, supplying water for industries and energy production. However, with the increasing global population and the relentless pursuit of development, the exploitation of river resources often occurs at the expense of the environment. This includes pollution, habitat destruction, and the neglect of sustainable practices, all of which lead to the degradation of river ecosystems (Srinivas et al., 2020).

The Narmada Parikrama is a significant religious pilgrimage that involves circumambulating the river. Pilgrims walk along the river's banks, visiting numerous temples and holy sites. The Narmada River is ecologically significant for its role in sustaining biodiversity, providing essential ecosystem services, and supporting human livelihoods. The region's distinct landscape and climate contribute to its exceptional biodiversity. The Narmada River hosts over 100 fish species, including several endemic and migratory species. Among the noteworthy species are the Mahseer, a highly sought-after game fish, and various catfish species, which play a crucial role in local fishing industries and the livelihoods of communities along the river (CIFRI, 2009; River at a Glance, 2024). Major threats and their impacts on the ecosystem are shown in Figures 1 and 2.

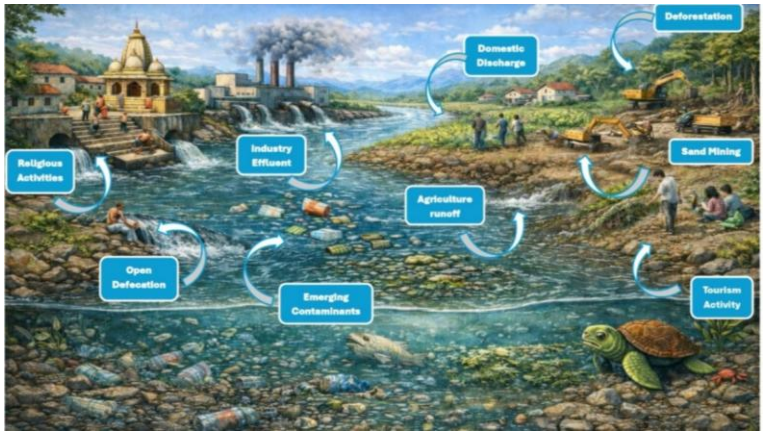


Figure 1: Major threats impacting the river ecosystem.

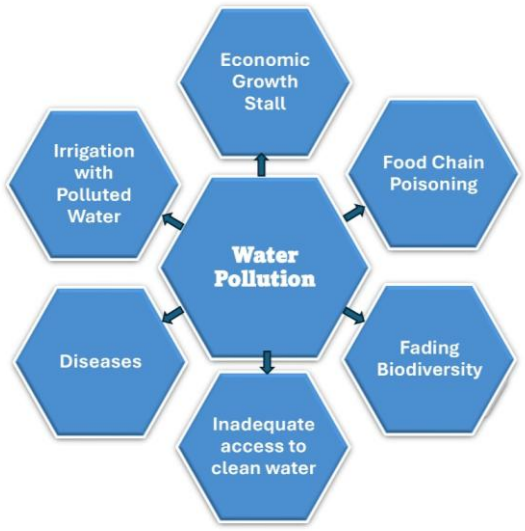


Figure 2: Impact of polluted water on aquatic life, human health, and the overall ecosystem.

3. Threats and Challenges for the Narmada River

3.1. Heavy/Toxic Metals

Heavy Metals have emerged as significant pollutants in river water, posing a considerable environmental challenge. These metals are non-biodegradable and pose potential risks to both aquatic organisms and humans, even at trace levels. Their presence in river water can result from natural processes, such as rock weathering and surface runoff, but the primary contributors are anthropogenic activities, particularly the discharge of untreated wastewater from industries. The bioaccumulation of heavy metals in aquatic organisms integrates them into the entire food chain (Arif et al., 2014).

In the study conducted by Gupta et al. (2025), sediment samples were analyzed from various districts in the upper basin of the Narmada River, including Amarkantak, Dindori, Mandla, and Jabalpur. The analysis focused on the distribution of heavy metals within these samples. The findings indicate that heavy metal concentrations are uneven across the upper basin, influenced by factors including varying input sources, geomorphology, and complex chemical reactions. The average concentrations of heavy metals are ranked in the following order: Fe > Al > Mn > Cr >Cu > Ni > Zn > Pb > Cd. The research highlighted that, based on the enrichment factor,

the Kapil -Dhara location in Amarkantak exhibited higher levels of heavy metals, along with the ghats in Dindori, Mandla, and Jabalpur, which also recorded significantly high concentrations. Notably, high levels of Mn, Cu, and Cr were observed at the Dindori station, and elevated Pb levels were found in Jabalpur. Ecological risk assessments indicate that sediment at the Dindori, Mandla, and Jabalpur sites is moderately to highly contaminated. The metals Mn, Cd, Pb, and Ni were identified as readily leachable due to their high mobility fractions, posing a significant risk to aquatic organisms and human health. This finding underscores the urgent need to properly monitor these heavy metals and implement measures to prevent their entry into the river system, which is vital to both the river ecosystem and human well-being. The author further suggests that the varying concentrations of heavy metals in the Narmada River are attributable to both natural (geogenic) and anthropogenic factors.

Station and reasons for the high concentration of heavy metals are summarized in Table 1. These include agricultural runoff, excessive use of pesticides and fertilizers, urban activities, mineral mining, untreated discharges, tourism, religious practices, and waste dumping.

Table 1: Sources of Heavy Metals at different locations **N1(Amarkantak) N2,3,4 (Dindori), N5,6,7 (Mandla) and N 8,9,10 (Jabalpur).** *Source:* https://doi.org/10.1080/03067319.2023.2299948

Heavy Metal	Highest Concentration Site (s)	Likely Source/Cause
Iron (Fe)	**N1** (Kapil Dhara)	Surface runoff from surrounding areas.
Aluminium (Al)	**N1** (Kapil Dhara)	Weathering of rocks like bauxite.
Manganese (Mn)	**N1, N2, N3, N4**	Agricultural runoff (N3) and domestic sewage (N4).
Chromium (Cr)	**N1** (Kapil Dhara) & **N4** (Jogitikariya)	Runoff and untreated domestic sewage from Dindori town.
Copper (Cu)	**N5** (Manode); also high at N1, N3, N4	Ritual wastes, sewage, and agricultural runoff.
Nickel (Ni)	**N3** (Ram Ghat)	Agricultural runoff.
Zinc (Zn)	**N1** (Kapil Dhara)	Surface runoff.
Lead (Pb)	**N9** (Tilwara Ghat)	**Diesel boats** used for tourists and recreational purposes.
Cadmium (Cd)	**N5** (Manode)	**Direct discharge of sewage**, atmospheric deposition, or agricultural runoff

A report by (CWC, 2019; NWDA, 2020; Sakshi Sharma, 2020) The elevated iron levels in the Narmada River exceed the limits set by the Bureau of Indian Standards (BIS). According to BIS, the acceptable limit for iron is 0.3 mg/L (300 µg/L). An analysis of trace and toxic metals at 18 water quality monitoring stations in the Narmada Basin from August 2018 to December 2020 revealed that only 2 stations met the BIS 10500-2012 standards for toxic metal concentrations. Conversely, eight stations exceeded permissible limits, attributed to the presence of two or more toxic metals, including arsenic (As), cadmium (Cd), lead (Pb), nickel (Ni), and fluorine (F). Regarding cadmium, the BIS recommends an acceptable concentration of 0.003 mg/L (3µg/L) in drinking water. In August 2019, a measurement at Narmadapuram recorded a cadmium concentration of 3.09 µg/L.

3.2 Microplastics

Microplastics are the other major challenge for the environment as they are very small in size, approximately less than 5 mm in size, but due to their accumulation in large amounts and their non-biodegradable nature, they are an emerging contaminant with high potential risks to the entire ecosystem. MPs are present in cosmetics, detergents, personal care products, and industrial products. Large plastics also generate microplastics. Major sources of microplastics include domestic, industrial, and untreated discharges, as well as fishing activities, which make rivers a source of microplastics (Horton et al., 2017).

Microplastics adhere to aquatic plants and act as carriers of heavy metals and harmful chemicals, ultimately affecting the entire food chain by disrupting river biodiversity. A notable study by Patidar et al., 2024 examined various locations in Maheshwar and Omkareshwar, both situated in the Middle Narmada Basin. The selected sites were based on their residential, tourism, and agricultural activities. The research revealed that microplastics were present in both water and sediment samples. Water samples showed a significant abundance of polyester, while polyvinyl chloride was predominant in the sediments. The findings indicated that the highest levels of pollution were in samples from residential areas, followed by those from tourism and agricultural regions. The presence of microplastics in the river suggests a concerning degree of pollution stemming from household sewage, single-use plastics, ritual waste, and fishing gear. This research uncovered high levels of microplastics in sediments, posing risks to benthic organisms. These polymers carry potential carcinogenic and toxic effects, severely impacting human health. Overall, microplastics significantly affect both the river ecosystem and public health.

4. Wastewater and Sanitation -Related Threats to the Narmada River

4.1. Open Defecation Free (ODF) System and Sanitation Gaps

ODF is defined by the Ministry of Drinking Water and Sanitation (MoDWS) as the termination of faecal–oral transmission, evidenced by the absence of visible faeces in the environment, in every household, and in public/community institutions, and by the use of a safe technology option for faecal disposal.

Open defecation has historically been most prevalent among the poorest citizens in the cities or the countryside. This practice amounts to tonnes of feces introduced daily into the environment, regularly exposing India's children to excrement through direct contact. The situation contributed to nearly 100,000 diarrhoeal deaths of children under five years in India. A tremendous achievement, only possible because of the Government's flagship programme, the *Swachh Bharat Mission* (SBM) (Clean India Campaign), led by the Prime Minister himself. UNICEF has been a proud partner of the *Swachh Bharat Mission*, which is now in its second phase of implementation. India has made rapid progress in eliminating open defecation nationwide. The number of people defecating in the open in India has reduced significantly by an estimated 450 million people. (UNICEF, 2026). Figure 3 shows the ODF rating system.



Figure 3: ODF rating system

All the cities along the Narmada riverbank are ODF++/ODF+ status, meaning clean toilets with proper processing and disposal of sludge. Amarkantak, Dindori, Madla, Narmadapuram, and Jabalpur in the upper basin, and Maheshwar, Omkareshwar, and Barwani in the Middle Narmada basin are ODF-free districts (Madhya Pradesh Pollution Control Board, 2025; SBM

2.0, 2026). Table 2 shows the current ODF status and data for T. Coliforms and F. Coliforms in the Narmada Basin district, and Table 3 presents the ODF data in the Lower Basin, Gujarat (GPCB, 2026).

Despite the status achieved in the cities of both the upper and Middle Basins, the data on faecal coliforms and total coliforms for the districts of the Narmada Basin show a different picture. As shown in Figures 4 and 5, biological contaminants are present, indicating contamination of the river water, which may be due to illegal, untreated discharge.

Table 2: Presents the ODF status and data of F. Coliform and T. Coliform.

(Upper and Middle Basin)

Source:(https://sbm.gov.in/sbmgdashboard/StatesDashboard.aspx),(https://www.mppcb.mp.gov.in/narmada-report16-17-17-18eng.aspx).

District	ODF Status/city-wise	Station Name/District-wise	F. Coliform Avg.	F. Coliform Max	T. Coliform Avg	T. Coliform Max
Anuppur	ODF+	River Narmada at Pushkar Dam, Amarkantak	1.8	1.8	60.01	120
Dindori	ODF++	Dindori River Narmada at D/S after mixing with the city's domestic waste, Dindori, Near Jogotikaria Ghat, Dist. Dindori	1.95	3.7	54.5	120
Mandla	ODF+	River Narmada at Mandla Near Shamshanghat.	8.65	34	53.8	130
Jabalpur	ODF++	Rever Narmada at Tilwaraghat	5.35	17	165.3	1600
Narsinghpur	ODF++	Barmanghat, 100, d/s. of Mainghat.Narsingpur	4.96	14	52.54	150
Narmadapuram	ODF++	River Narmada at Shaha Ganj Rest House	44.58	138	190.33	240
Khandwa	ODF++	Narmada River at Punasa Dam	4.5	18	40	48
Omkareshwar	ODF++	At Omkarewhwar d/s.	1.8	1.8	43	49
Mandleshwar	ODF++	At Mandleshwar d/s	1.8	1.8	40.167	47
Maheshwar	ODF++	At Shahastradhara (Jalkoti)	1.8	1.8	101.285	465
Alirajpur	ODF++	At Kakrana, the interstate boundary, Alirajpur.	1.8	1.8	41.5	49

Table 3: Presents the ODF status (Lower Basin)
Source: **https://sbmurban.org/swachh-certification-2024-25-result**, https://gpcb.gujarat.gov.in/ (2024-2025)

District	ODF Status	Location	F. Coliform (MPN/ 100ml)	T. Coliform (MPN/100ml)
Narmada	ODF++	NTPC Zanor	40	93
		Zadeshwar	61	140
		Garudeshwar	40	92
Bharuch	Water +	-	-	-
Chota Udepur	Non-ODF++	-	-	-

As observed from the data, cities in the Narmada Basin, such as Annupur, Dindori, Mandla, Jabalpur, Omkareshwar, and Maheshwar, are ODF++ status, but faecal and total coliform data show high values in the monsoon and post-monsoon months, as per MPPCB 2024-2025 data. This alarming data indicates a high risk of contamination in the Narmada River. There could be many reasons for this biological contamination in the river water.

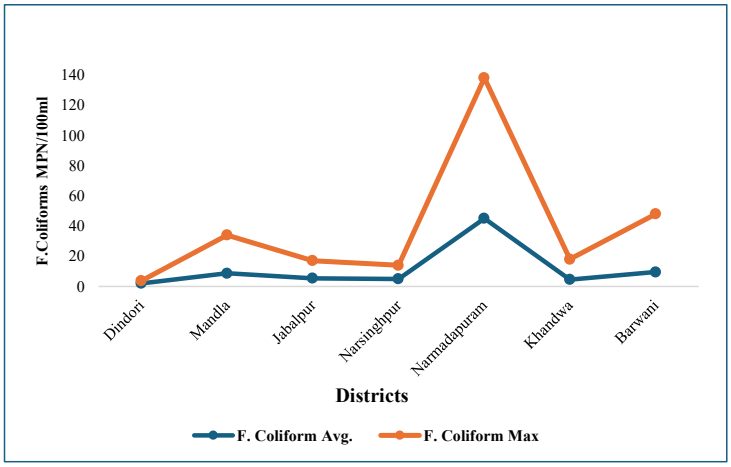


Figure 4: F. Coliform data of Districts in the Narmada Basin

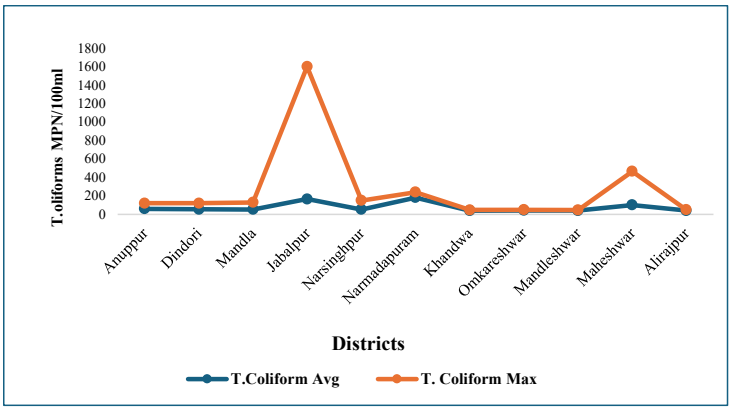


Figure 5: T. Coliform data of Districts in the Narmada Basin

According to the WHO guidelines on sanitation (2018), many challenges remain in achieving proper sanitation and maintaining ODF status. These challenges include inadequate sanitation facilities in rural areas, poor management, public awareness, and climate change.

5. Sediment Load Impact on the Narmada River Water Quality

There are various point and non-point sources of sediment in the river. Point sources of sediments are fewer compared to non-point sources. Point sources of sediment include construction- and dam-related discharges, which are specific and identifiable locations from which sediments enter the river. In addition to construction and dam sites as sources, sediments from gravel mining, industrial areas, and domestic waste also serve as point sources into the river, as shown in Figure 6.



Figure 6: Presence of solid and liquid waste on the Narmada River at Narmadapuram(Source, cNarmada Team)

Besides specific sources of sediment, a significant portion comes from non-point sources that encompass broad, unspecified areas rather than a single location. Most sediment loads surge during the monsoon season, accounting for approximately 70% of the annual sediment load. Activities such as farming on sloped terrains, deforestation, land use changes, and disturbances in steep upstream areas significantly increase sediment delivery to rivers during heavy rainfall. This wash-off of topsoil from agricultural lands, forests, and hilly areas not only increases sediment load in the river but also disrupts the ecological balance, leading to further environmental challenges (Rahi et al., 2024, Nutrient and Sediment Load Report, 2025).

Anthropogenic activities, coupled with nutrient sediment load and domestic and industrial sewage, are negatively impacting the ecological health of the river (Sarsaiya et al., 2025). One study has shown that elevated nutrient levels, such as nitrogen and phosphorus, have led to increased growth of certain macrophyte species, which serve as bioindicators of water quality. These macrophytes thrive in areas with high organic load. Specific locations, including Kabir Sarovar and Kharighat, have demonstrated the presence of these macrophytes. Water samples from these sites indicate the accumulation of pollutants and organic matter, raising concerns that necessitate immediate action to control pollution (Saini et al., 2025).

6. Agriculture Practices

In the Narmada basin, various farming methods are used. In the upper areas, people rely on rain-fed farming, meaning they depend on rainfall for irrigation. In areas with canals, farmers practice intensive irrigated agriculture, using canal water to grow their crops. These different farming methods have a big impact on how nutrients like nitrogen (N), phosphorus (P), and potassium (K) cycle through the soil. Different cropping patterns, irrigation practices, crop residue management, and riverbank farming collectively affect river water quality (Central Ground Water Board, 2020).

In the Narmada basin, soybean-wheat and sugarcane-based agriculture are prevalent. Soybeans are mainly grown as a legume crop that fixes nitrogen, but after residual removal or burning, more than 90% of the nitrogen is lost, reducing soil organic matter. This makes farmers rely on fertilizers to improve crop yields. And these fertilizers enter the river water, elevating its nitrate levels. Subsequently, poor residue management results in high nutrient losses and reduced biofertilizer inputs, leading to low soil organic carbon and diminished microbial activity. Another important factor in flood irrigation patterns is the widespread use of this method in the Narmada Basin, which is a major driver of surface runoff and the leaching of nitrogen and phosphorus in drainage, tributaries, and river stretches. In the Narmada basin, the drip and sprinkle irrigation percentages are low, despite their advantages (CPCB Water Quality Reports (2022), Singh et al. (2021), Central Groundwater Board (2020) The broad adoption of precision irrigation systems could significantly reduce nutrient runoff, enhance nutrient uptake, and support sustainable nutrient cycling throughout the basin (Kansara & Lakshmi, 2021).

Another important thing is farming in the Riverbank and floodplain areas, which are fertile for horticulture, but the heavy use of fertilizers leads to nutrient runoff into the river and its tributaries, such as the Tawa and the Sher. This riverbank farming is one of the key reasons for rising nutrient levels in the river, which leads to eutrophication (Narmada River Basin Agriculture Profile Report, 2025; Nutrient and Sediment Load Report, 2025) .

The extensive use of nitrogen- and phosphate-based fertilizers has contributed to water pollution within the Narmada basin. Water quality samples collected from the Khandwa district indicate elevated concentrations of nitrates and phosphates, which are primary contributors to eutrophication and algal blooms in the river, thereby reducing oxygen levels and affecting aquatic life. Additionally, sediment samples collected from agricultural zones such as Sehore

and Barwani reveal the presence of heavy metals, likely due to prolonged and consistent use of phosphate fertilizers and pesticides (River Water Quality Monitoring Report –CPCB, 2019).

Commented [MJ1]: citations

The presence of heavy metals and chemicals can lead to health issues such as methemoglobinemia, also known as "blue baby syndrome," in infants. Additionally, exposure to these substances may increase the risk of cancer and neurological disorders. The report also mentioned that extensive groundwater dependence in the districts of Khandwa and Narmadapuram will affect the Narmada River, as they are interconnected (CGWB, 2022)

Riparian vegetation on riverbanks is also dependent on the river and groundwater. And this riparian vegetation adjusts to the changes in river water levels and soil moisture (Lambs, 2000).

7. Sand Mining

Extensive sand mining reduces the deposition of nutrient-rich alluvial sediments on floodplains, lowering soil fertility, increasing dependence on chemical fertilizers, and accelerating riverbank erosion.

Sand consumption in Madhya Pradesh, central India, in 2017 was recorded at approximately 56.80 million metric tonnes and is projected to increase to 90.53 million metric tonnes by 2025. About 1266 sand mines are located in Madhya Pradesh, of which around 13% are located along the Narmada River (Upadhyay, 2019). The study at Sandia and Narmadapuram gauge stations highlighted the effects of Sand mining. Sand mining has led to riverbed incision, increased flow velocity, and conveyance capacity, resulting in higher erosive power and more frequent floods. These changes pose risks to regional biodiversity, water availability, and geomorphic equilibrium. The research highlights the significant hydrogeomorphic impacts of sand mining, particularly in the upstream region, and emphasizes the need for sustainable mining practices to protect river systems and local livelihoods. This study provides a framework for understanding and generalizing the impacts of sand mining on alluvial rivers, emphasizing the importance of sustainable resource extraction to prevent irreversible environmental degradation (Singh et al., 2026). Figure 7 below shows illegal sand mining in the Narmadapuram region.



Figure 7: Sand Mining location of Narmadapuram ***Source:***https://mpcg.ndtv.in/madhya-pradesh-news/narmadapuram-administration-took-action-against-illegal-sand-mining-in-narmada-river-seized-five-boats-stealing-sand-4996878

8. Construction of Dams

Environmental flows, or "e-flows," are the water levels and flow patterns that must be maintained in rivers to support healthy ecosystems. Managing these flows becomes especially important in river basins where dams and other water projects alter the natural water regime. (Vengadesan et al., 2023).

The Bargi Dam on the Narmada River in the Jabalpur District of Madhya Pradesh is one such example where changes in water flow have had significant ecological impacts. Dams like Bargi have multiple purposes – they provide water for drinking, irrigation, and power generation. However, when a dam is built, it alters downstream water flow. This alteration can lead to a variety of problems, including reduced water quality, loss of fish habitat, and disrupted seasonal flooding (Socio-environment of Narmada Basin, 2025).

The construction of large dams, such as the Sardar Sarovar and Narmada Sagar dams on the Narmada River, has resulted in significant freshwater storage within the reservoirs. Consequently, the river's downstream stretches are receiving less freshwater. According to the Narmada Tribunal report, after 30 years, this reduction in freshwater flow is projected to reach 72%, which is a substantial decrease that will impact the river's ecosystem. If insufficient freshwater reaches the estuary, the natural balance between seawater and river water will be disrupted. The absence of freshwater dilutes seawater salinity, which is crucial for migratory fish species (CIFRI, 2009).

As large dams continue to be constructed and the amount of freshwater flowing into downstream sections and ultimately into the sea decreases, the estuary is likely to become more saline as tidal waters penetrate further into the river. This imbalance between freshwater and saline water will adversely affect migratory species such as *Tenualosa Ilisha* (hilsa) and *Macrobrachium Rosenbergii*, which rely on a balanced mix of fresh and saline waters for breeding and survival. As salinity increases and habitats change, fish populations are expected to decline, leading to reduced fish yields and a significant transformation of the estuarine ecosystem. This shift will ultimately impact the livelihoods of downstream fishing communities (Dasgupta et al., 2017).

9. Pollution Load affecting the Narmada River (***Source:*** ***https://www.mppcb.mp.gov.in/District-Environment-Plans.aspx*****)2025**

Major sources of pollution in the Narmada River include sewage generated in the districts and industrial wastewater. Many sewage-generating cities and industrial discharge facilities lack adequate wastewater treatment facilities, such as STPs and CETPs, and, as a result, untreated wastewater is entering the Narmada River directly or indirectly**.** Many cities in the Narmada basin have operational STP status, but most have limited treatment capacity, resulting in a significant processing gap.

Inadequate and sub-optimal functioning of Sewage Treatment Plants

One of the major threats to the Narmada River is the discharge of untreated sewage and industrial wastewater. Analysis of the data reveals that sewage wastewater generation exceeds treatment capacity, largely due to the insufficient number of sewage treatment plants (STPs) in the basin. While some STPs are proposed or under construction, cities along the riverbanks, particularly Narmadapuram, Balaghat, Alirajpur, and Umaria, lack operational STPs. Additionally, urban areas such as Jabalpur, Khandwa, Khargone, and Chhindwara suffer from inadequate sewage treatment (MPPCB, 2025). Figures 8 and 9 show the existing sewage treatment infrastructure and processing gaps in the Narmada basin districts, as well as total sewage generation and STP capacity. Figure 10 shows the Total Sewage Generation and STP capacity in the Upper and Middle Narmada Basin.

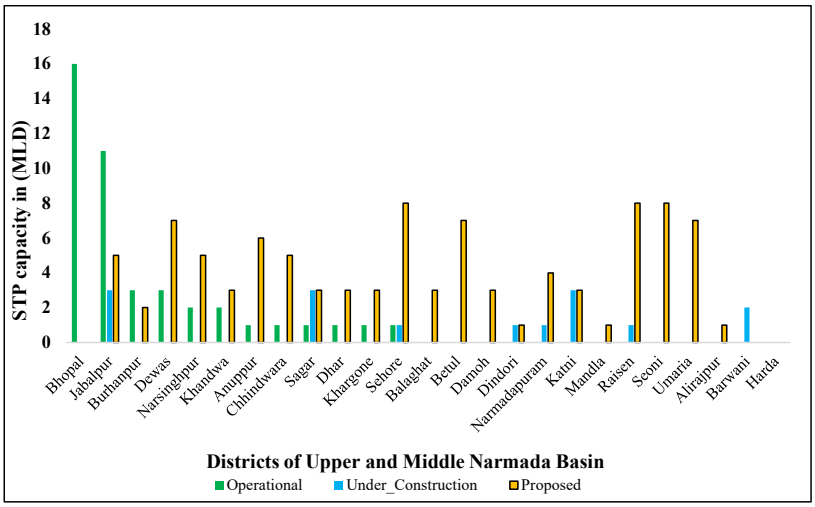


Figure 8: STP status of the Districts of the Upper and Middle Narmada Basin

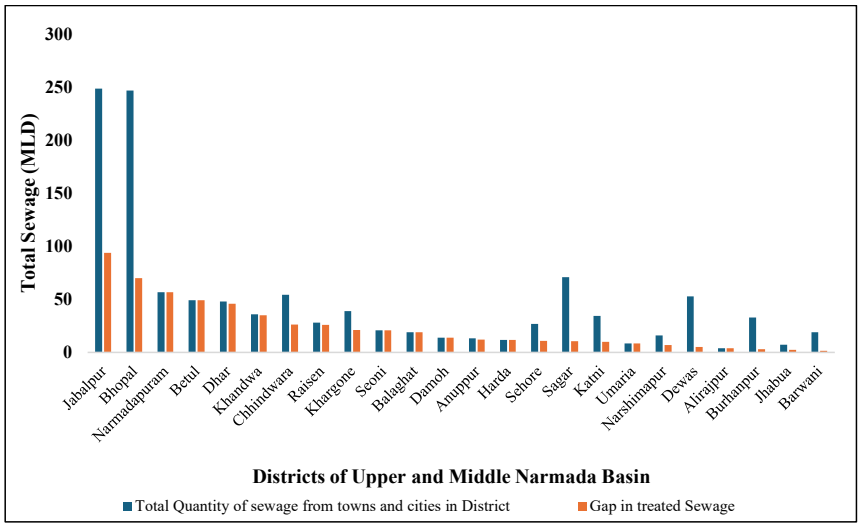


Figure 9: Total Quantity of Sewage Generated and The Processing Gap

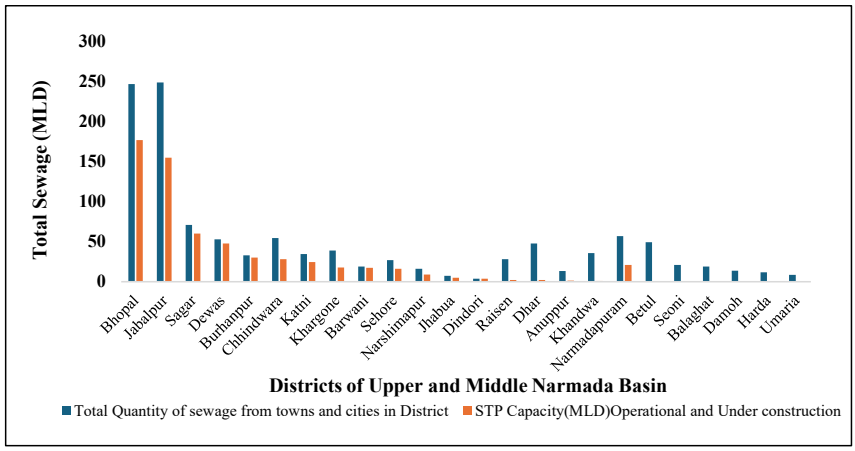


Figure 10: Total Sewage Generation and STP Capacity

It has been observed that the total quantity of sewage generated in the cities and towns of the Narmada basin districts significantly exceeds the capacity for proper treatment. This imbalance indicates that more sewage is being generated than can be processed, highlighting a critical issue: the discharge of untreated wastewater into the Narmada River. This disparity between wastewater generation and processing highlights a significant lack of technology and infrastructure in the cities of the Narmada basin. The data indicate that, due to insufficient numbers of STPs, untreated wastewater is discharged into drains and river stretches, often flowing directly into the river.

Inadequate and sub-optimal functioning of Common Effluent Treatment Plants. Industrial wastewater generation in the Narmada Basin varies significantly across districts, reflecting the uneven distribution of industries. The data in Figure 11 reveal that while some districts, such as Betul, Jabalpur, Chhindwara, Dhar, Raisen, and Khargone, have substantial industrial discharge, others report negligible or no industrial wastewater.

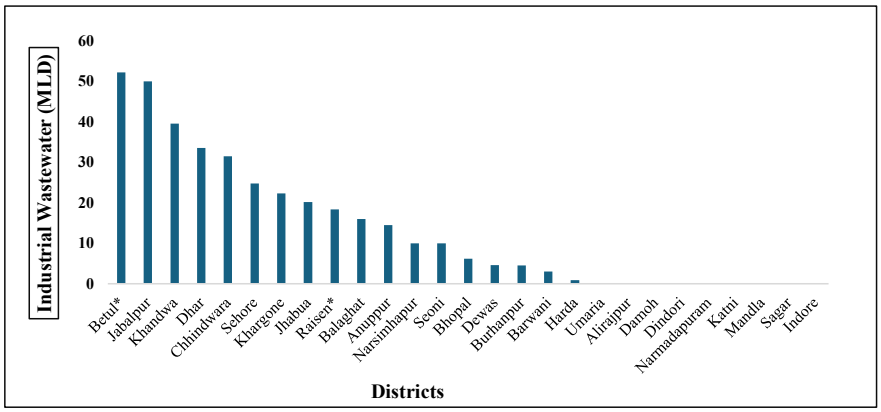


Figure 11: Industrial Wastewater Generation in the Narmada Basin

Source: https://www.mppcb.mp.gov.in/District-Environment-Plans.aspx

According to MPPCB data (2025), some districts have industrial discharges and are reported to meet environmental standards, while other districts report negligible discharge.

The Centralized Effluent Treatment Plant (CETP) located at Sindhi Basti in Burhanpur has a capacity of 4.5 MLD. Additionally, there is a second CETP in Jabalpur with a combined capacity of 25 MLD.

Currently, there are three existing CETPs with a total capacity of 8.7 MLD located in Bhopal, Indore, and Burhanpur. Furthermore, two newly constructed CETPs, each with a capacity of 0.6 MLD, have been established in Indore and Jhabua. Most districts lack common treatment facilities, forcing industries to rely on individual Effluent Treatment Plants (ETPs). While compliance with standards is generally reported as satisfactory in many districts, the absence of CETPs raises concerns about long-term monitoring, enforcement, and sustainability of wastewater treatment.

10. Religious and tourism activities

A study conducted at Riddhnath Ghat in Handiya Village, located in Harda District, observed increases in various physical parameters of water quality following mass bathing on multiple occasions. These religious activities and the influx of bathers led to significant variations in key water quality indicators, including dissolved oxygen (DO), biochemical oxygen demand (BOD), total dissolved solids (TDS), turbidity, and hardness. Such changes

are critical for assessing river water quality and the overall ecosystem. The pollution stemming from these activities not only degrades water quality but may also introduce pathogens that can cause waterborne diseases, further contaminate the river, and pose risks to human health. These effects not only impact a specific stretch of the river but also have broader implications for aquatic life and public health (Rathore et al., 2023).

The ghats in Mandla district have been found to have poor water quality. This is mostly because many pilgrims come to perform rituals and take baths there. Similarly, the water quality at the ghats in Narmadapuram, such as Sethani and Budhini Ghat, is poor due to the large number of visitors for religious activities (Gupta et al., 2020).

Rising tourism reliance in the Narmada Basin due to the popularization of hotspots along the Narmada River, such as Bhedaghat and Omkareshwar in Madhya Pradesh, and the Statue of Unity and Sardar Sarovar Dam in Gujarat, attracts millions of visitors annually, leading to pollution and environmental degradation, and is expected to put the river's health at risk in the future (Narmada River Basin River-People Connect Report, 2025) .

The practice of open grazing, coupled with deforestation driven by agricultural expansion, particularly shifting cultivation, has exacerbated soil erosion in the upper and middle catchments of the Narmada Basin, specifically in districts such as Mandla, Dindori, Chhindwara, and Betul. This increased erosion not only strips the land of its fertile topsoil but also contributes to river sedimentation, further harming aquatic habitats (Ahirwar et al., 2019; Meshram et al., 2022).

Rapid urbanization is also a critical issue, as cities like Jabalpur, Hoshangabad, and Bharuch have transformed once-fertile agricultural and forest lands into concrete jungles. This transition has disrupted essential soil–crop–vegetation linkages, leading to increased runoff and the introduction of substantial nutrient loads into water bodies from untreated sewage and industrial waste. This influx contributes to localized eutrophication, resulting in harmful algal blooms that threaten aquatic life and degrade water quality(Kansara & Lakshmi, 2021).

In the upper basin districts, such as Mandla, Dindori, and Balaghat, the relentless deforestation for agriculture and infrastructure projects has decreased organic matter inputs into the soil. This reduction weakens the soil's ability to buffer nutrients, making it more susceptible to erosion and increasing downstream nutrient loading. Consequently, the ecological balance is further compromised, posing a severe threat to the sustainability of both

the river ecosystem and the broader agricultural landscape (Nutrient and Sediment Load Report, 2025).

11. Conclusion

The Narmada River is a vital lifeline for central and western India. Its health is essential to the entire ecosystem, providing a variety of benefits, including access to drinking water, irrigation, hydroelectric power, and cultural heritage. This report highlights the major issues impacting the river's health, including water quality, flow, aquatic life, and riparian and floodplain vegetation.

These challenges stem from activities such as deforestation, sand mining, infrastructure development along the river, and pollutants from domestic and industrial effluents, which contribute to the presence of emerging contaminants. Open defecation, various religious practices, waste dumping, and boating further exacerbate the situation. The generation of contaminants like heavy metals and microplastics is particularly concerning, as these substances are difficult to degrade and pose risks to both aquatic life and human health.

Collaborative efforts among policymakers, researchers, and stakeholders are essential to address these challenges and restore the ecological balance of the Narmada River while supporting socioeconomic progress in the region.

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